CBSE Class 10 Mathematics (Standard) – 2020

Question Paper Code: 30/1/2 | Set 2

SECTION A

Multiple Choice Questions (1 Mark Each)

Q1. In the given figure, PA is a tangent from an external point P to a circle with centre O. If $\angle POB = 115^\circ$, then $\angle APO$ is equal to:

(A) 25°
(B) 65°
(C) 90°
(D) 35°

Q2. A piece of wire 20 cm long is bent into the form of an arc of a circle of radius ($\frac{60}{\pi}$) cm. The angle subtended by the arc at the centre of the circle is:

(A) 30°
(B) 60°
(C) 90°
(D) 50°

Q3. Three numbers in AP have the sum 30. What is its middle term?

(A) 4
(B) 10
(C) 16
(D) 8

Q4. An arc of a circle is of length (5π) cm and the sector it bounds has an area of (20π) cm^2. Find the radius.

(A) 10 cm
(B) 1 cm
(C) 5 cm
(D) 8 cm

Q5. If ($x=1$) and ($y=2$) is a solution of

$$2x-3y+a=0$$

and

[
2x+3y-b=0
]

then:

(A) (a=2b)
(B) (2a=b)
(C) (a+2b=0)
(D) (2a+b=0)

Q6. Two polynomials are shown in the graph. The number of distinct zeroes of both polynomials is:

(A) 3
(B) 5
(C) 2
(D) 4

Q7. If ($\alpha+\beta=90^\circ$) and ($\alpha=2\beta$), then

[
$\cos^2\alpha+\sin^2\beta$
]

is equal to:

(A) 0
(B) ($\frac12$)
(C) 1
(D) 2

Q8. A card is selected at random from a deck of 52 cards. The probability that it is a red face card is:

(A) ($\frac{3}{13}$)

(B) ($\frac{2}{13}$)

(C) ($\frac12$)

(D) ($\frac{3}{26}$)

Q9. If α and β are the zeroes of the polynomial

[
$3x^2+6x+k$
]

such that

$$
\alpha+\beta+\alpha\beta=-\frac23
$$

then the value of k is:

(A) −8
(B) 8
(C) −4
(D) 4

Q10. Find the value of

$$
\tan^2\theta-\left(\frac1{\cos\theta}\times\sec\theta\right)
$$

(A) 1
(B) 0
(C) −1
(D) 2

Q11. Which of the following is a rational number between $\sqrt3$ and $\sqrt5$?

(A) 1.4142387954012...
(B) $2.32\overline6$
(C) π
(D) 1.857142

Q12. If HCF(98, 28) = m and LCM(98, 28) = n, then $n-7m$ equals:

(A) 0
(B) 28
(C) 98
(D) 198

Q13. If the length of a chord of a circle is equal to its radius, then the angle subtended by the chord at the centre is:

(A) 60°
(B) 30°
(C) 120°
(D) 90°

Q14. The greatest number which divides 70 and 125 leaving remainders 5 and 8 respectively is:

(A) 13
(B) 65
(C) 875
(D) 1750

Q15. A ladder 14 m long leans against a wall. If the foot of the ladder is 7 m from the wall, then the angle of elevation is:

(A) 15°
(B) 30°
(C) 45°
(D) 60°

Q16. In triangles ABC and DEF, if

$\angle B = \angle E$, $\angle F = \angle C$ and $AB = 3DE$,

then the triangles are:

(A) Congruent but not similar
(B) Congruent as well as similar
(C) Neither congruent nor similar
(D) Similar but not congruent

Q17. The midpoint of the line segment joining

$P(-4, 5)$ and $Q(4, 6)$

lies on:

(A) x-axis
(B) y-axis
(C) Origin
(D) Neither x-axis nor y-axis

Q18. Mode and Mean of a data are 15x and 18x respectively. The Median is:

(A) x
(B) 11x
(C) 17x
(D) 34x

Assertion–Reason Questions

Q19.

Assertion (A): If we join two hemispheres of the same radius along their bases, we get a sphere.

Reason (R): Total surface area of a sphere of radius r is ($3\pi r^2$).

Q20.

Assertion (A): The probability of selecting a number at random from the numbers 1 to 20 is 1.

Reason (R): For any event E, if P(E) = 1, then E is called a sure event.

SECTION B

Very Short Answer Questions (2 Marks Each)

Q21. If the zeroes of the polynomial

$$x^2+ax+b$$

are in the ratio 3 : 4, prove that

$$12a^2=49b$$

Q22. A person stands at point P outside a circular ground at a distance of 26 m from its centre. His distances from points A and B on the ground are 10 m each (PA and PB are tangents to the circle). Find the radius of the circular ground.

Q23. (a) If

$$\triangle ABC \sim \triangle PQR$$

where

AB = 6 cm, BC = 4 cm, AC = 8 cm and PR = 6 cm,

find the value of (PQ + QR).

OR

(b) In the given figure, if

$$
\frac{QR}{QS}=\frac{QT}{PR}
$$

and ∠1 = ∠2, show that

$$
\triangle PQS \sim \triangle TQR
$$

Q24. (a) If

$$
x\cos60^\circ+y\cos0^\circ+\sin30^\circ-\cot45^\circ=5
$$

find $(x+2y)$.

OR

(b) Evaluate:

$$
\frac{\tan^2 60^\circ}
{\sin^2 60^\circ+\cos^2 30^\circ}
$$

Q25. The coordinates of the centre of a circle are

$$
(2a,\ a-7)
$$

If the circle passes through (11, −9) and has diameter $(10\sqrt2)$ units, find the value(s) of (a).

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SECTION C

Short Answer Questions (3 Marks Each)

Q26.

If the radii of the bases of a cylinder and a cone are in the ratio 3 : 4 and their heights are in the ratio 2 : 3, find the ratio of their volumes.

Q27.

Three sets of Physics, Chemistry and Mathematics books are to be stacked in such a way that all the books are stored subject-wise and the height of each stack is the same.

The number of books are:

- Physics = 144
- Chemistry = 180
- Mathematics = 192

Assuming all books have equal thickness, determine the number of stacks of each subject.

Q28.

Two dice are thrown simultaneously.

Find the probability that the difference of the numbers obtained on the two dice is 2.

Q29. (a)

Prove that:

[
\frac{\tan\theta}{1-\cot\theta}
+
\frac{\cot\theta}{1-\tan\theta}

1+\sec\theta\cosec\theta
]

OR

Q29. (b)

Prove that:

[
\frac{\sin A+\cos A}{\sin A-\cos A}
+
\frac{\sin A-\cos A}{\sin A+\cos A}

\frac{2}{2\sin^2A-1}
]

Q30. (a)

In the given figure, O is the centre of the circle and BCD is tangent to the circle at C.

Prove that:

[
\angle BAC+\angle ACD=90^\circ
]

OR

Q30. (b)

Prove that opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle.

Q31.

Find the ratio in which the y-axis divides the line segment joining the points:

[
(5,-6)\ \text{and}\ (-1,-4)
]

Also find the point of intersection.

SECTION D

Long Answer Questions (5 Marks Each)

Q32. (a)

The diagonal BD of a parallelogram ABCD intersects the line segment AE at point F, where E is any point on side BC.

Prove that:

[
DF \times EF = FB \times FA
]

OR

Q32. (b)

In ΔABC, if

$$
AD \perp BC
$$

and

$$
AD^2 = BD \times DC
$$

prove that:

$$
\angle BAC = 90^\circ
$$

Q33.

The following frequency distribution gives the monthly consumption of electricity of 68 consumers of a locality.

Monthly Consumption (Units)	**Number of Consumers**
65 – 85	4
85 – 105	5
105 – 125	13
125 – 145	20
145 – 165	14
165 – 185	8
185 – 205	4

Find:

1. Mean
2. Mode

Q34.

Vijay invested certain amounts in two schemes A and B which offer interest at the rate of 8% p.a. and 9% p.a. respectively.

He received ₹1860 as total annual interest.

If the invested amounts are interchanged, he would receive ₹20 more as annual interest.

Find the amount invested in each scheme.

Q35. (a)

A two-digit number is such that the product of its digits is 12.

When 36 is added to the number, its digits interchange their places.

Find the number.

OR

Q35. (b)

A student scored a total of 32 marks in Mathematics and Science.

Had he scored 2 marks less in Science and 4 marks more in Mathematics, the product of his marks would have been 253.

Find his marks in the two subjects.

SECTION E

Case Study Based Questions (4 Marks Each)

Q36. Case Study – Brooch

A brooch is a decorative piece often worn on clothing. A circular brooch of diameter 35 mm contains a design made using sectors and arcs.

Based on the given figure, answer the following:

1. Find the radius of the brooch.
2. Find ∠ACB.
3. Find the area of the shaded region.
4. Internal choice question based on the same figure.

Q37. Case Study – Statistics

A survey involving students' performance was conducted and the data was represented graphically.

Based on the given data answer the questions related to:

1. Mean
2. Median
3. Mode
4. Interpretation of statistical observations

Q38. Case Study – Probability

An experiment involving random selection was conducted.

Based on the given information answer questions related to:

1. Sample space
2. Favourable outcomes
3. Probability calculation
4. Related application-based question

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Total Questions: 38